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TIGIT affects CAR-NK cell effector function in the solid tumor microenvironment by modulating immune synapse strength

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Abstract

Therapies using natural killer (NK) cells that express chimeric antigen receptors (CAR-NKs) have been successfully employed against hematological malignancies. However, solid tumors resist CAR-NKs partly by enriching tumor microenvironments with ligands for NK cell inhibitory receptors. Although the NK inhibitory receptor TIGIT has been implicated in impaired anti-tumor activity of endogenous NK cells, the consequences of TIGIT expression on engineered CAR-NKs has not been explored. To address this gap, we compared TIGIT-expressing and TIGIT-deleted human CAR-NKs targeting the GD2 solid tumor antigen in tumor immune microenvironment (TiME) co-cultures and *in vivo* TiME xenografts designed to mimic the immunosuppressive environment of solid tumors. TIGIT-deleted GD2.CAR-NKs exhibited antitumor activity, expanded, and persisted within TIGIT ligand-enriched solid tumor environments, while TIGIT-expressing CAR-NKs did not. Mechanistic experiments revealed that the improved tumor control resulting from TIGIT loss on CAR-NKs was not dependent on DNAM-1 activation or enhanced cytotoxic potential, but rather on downregulation of cell adhesion molecules, weakened cell avidity, and reduced synapse contact duration that, in concert, improved serial killing and allowed more efficient tumor destruction. Our study highlights a non-canonical role for TIGIT in modulating CAR-NK activity that may guide strategies to overcome inhibitory NK receptors like TIGIT and improve efficacy of CAR-NKs against solid tumors.

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Keywords

CAR-NK cells; TIGIT; Immune Synapse; Solid Tumors; Tumor Microenvironment

INTRODUCTION

Natural killer (NK) cells are cytotoxic lymphocytes of the innate immune system. Their cytotoxicity against target cells is governed by a balance of signals from germline-encoded activating and inhibitory receptors expressed on the NK cell surface (1). In contrast to adoptive T cell-based therapies for cancer, adoptive transfer of allogeneic NK cells does not result in graft-vs-host disease or cytokine release syndrome (2,3). In fact, therapy with allogeneic NK cells engineered to express a CD19-targeting chimeric antigen receptor (CAR) and interleukin (IL)-15 resulted in improved outcomes in patients with relapsed or refractory B-cell malignancies without mediating the typical CAR-driven toxicities seen with T cells (4,5). While CAR-expressing natural killer cells (CAR-NKs) have been successful in hematological malignancies, their efficacy has been limited against solid tumors. This is partly due to a highly suppressive solid tumor immune microenvironment (TiME) that includes inhibitory myeloid cells such as M2 macrophages (M2-TAMs) and myeloid-derived suppressor cells (MDSCs) which express ligands against inhibitory receptors on infiltrating NK cells resulting in decreased effector function (6).

T-cell immunoreceptor with Ig and immunoreceptor tyrosine-based inhibitory motif domains (TIGIT) is an inhibitory receptor on NK cells and T cells whose ligands are upregulated in multiple solid tumors (7–9). TIGIT mediates its suppressive effect on NK cells by binding to its most abundant ligands, CD155 and CD112, overexpressed on malignant cells as well as M2-TAMs and MDSCs. This triggers inhibitory signaling in NK cells through TIGIT's intracellular immunoreceptor tyrosine-based inhibitory motif (ITIM) and immunoglobulin tail tyrosine (ITT)-like domains (10). Given that TIGIT impairs cytotoxicity by endogenous NK cells and its overexpression is correlated with poor prognosis in solid tumor patients (7,11,12), the TIGIT-ligand axis likely represents a powerful barrier to CAR-NK efficacy within solid TiMEs. While the role of TIGIT in endogenous solid tumor-infiltrated NK cell activity has been studied, its effect on the behavior of CAR-NKs in these environments has not been reported. Addressing this gap in knowledge is clinically relevant as understanding mechanisms by which TIGIT impairs CAR-NK functions would allow design of biology-driven strategies to overcome TIGIT inhibition and improve efficacy of CAR-NK products designed for solid tumor anti-cancer therapy.

To interrogate the effect of persistent TIGIT engagement on CAR-NK anti-tumor responses specifically within the TiME, we compared a solid tumor-targeting CAR-NK with efficient and stable TIGIT knockout to TIGIT-expressing CAR-NKs in a TiME co-culture and an *in vivo* TiME xenograft model designed to mimic the immunosuppressive environment of solid tumors. We show herein that TIGIT deletion enhanced CAR-NK proliferation and serial killing, conferring tumor control that extended the survival of mice bearing TiME xenografts. The enhanced CAR-NK functionality was not due to improvements in canonical signaling pathways normally inhibited ty TIGIT, but instead, to a role for TIGIT in modulating immune synapse strength in CAR-NKs upon engagement with tumor targets. Our findings reveal a role for TIGIT in affecting anti-tumor function in CAR-NKs with implications in redefining the conventional thinking on the role of endogenous inhibitory receptors in lymphocytes engineered to express CARs.

METHODS

Cytokines and cell lines.

Recombinant human interleukin IL-2 was obtained from National Cancer Institute Biologic Resources Branch (Frederick, MD). Recombinant human IL-4 (200–04-100UG), IL-6 (200–06-100UG), M-CSF (300–25-100UG), GM-CSF (300–03-100UG), and IL-15 (200–15-100UG) were purchased from Peprotech (Rocky Hill, NJ, USA). The human neuroblastoma cell lines CHLA255 (RRID:CVCL_AQ27, obtained 2021) and LA-N-1 (RRID:CVCL_1827, obtained 2017) were purchased from American Type Culture Collection (Manassas, VA, USA) and cultured in RPMI-1640 culture medium (Hyclone SH30096.01) supplemented with 2 mM L-glutamine (Gibco-BRL Cat# 35050061) and 10% FBS (Gibco-BRL Cat# A5670701). The human lung cancer cell line H1650 (RRID: CVCL_1483, obtained 2024) was purchased from American Type Culture Collection and cultured in complete-DMEM culture medium composed of DMEM medium (Hyclone Cat# SH30285.01) supplemented with 2 mM L-glutamine and 10% FBS. The human CML cell line K562 (RRID:CVCL_0004, obtained 2011) was purchased from American Type Culture Collection and cultured in complete-RPMI culture medium composed of RPMI-1640 medium (Hyclone) supplemented with 2 mM L-glutamine and 10% FBS. A modified version of parental K562 cells, genetically modified to express a membrane-bound version of IL-15 and 41BB-ligand, K562-mb15–41BB-L, was kindly provided by Dr. Dario Campana (National University of Singapore), obtained 2011. A modified version of parental K562, genetically modified to express CD80, CD86, and 41BB-L, K562-mb15-CS, was kindly provided by Dr. Cliona Rooney (Baylor College of Medicine), obtained 2020. The human epithelial cell line HEK293T (RRID:CVCL_0063) purchased from American Type Culture Collection (Manassas, VA, USA) and cultured in DMEM culture medium supplemented with 2 mM L-glutamine (Gibco-BRL) and 10% FBS (Gibco-BRL), obtained 2015. All cell lines were verified by either genetic or flow cytometry-based methods and tested for mycoplasma contamination monthly via MycoAlert (Lonza Cat# LT07–318) mycoplasma enzyme detection kit. All cell lines were used within one month of thawing from early-passage (< 3 passages of original vial) lots.

CAR-encoding retroviral vectors.

The construction of the SFG-retroviral vector encoding GD2.CAR.41BB.ζ, as shown in Supplementary Fig. S1A, was previously described (13,14). The HER2-specific scFv (FRP5)-based CAR transgenes were described previously (15). RD114-speudotyped viral particles were produced by transient transfection in 293T cells, as previously described (16).

Antibodies and flow cytometry.

The following antibodies were used in this study; CD56 (BD Biosciences Cat# 555516, RRID:AB_395906), CD3 (BD Biosciences Cat# 345763, RRID:AB_2811220, BioLegend

Cat# 344834, RRID:AB_2565675), TIGIT (BioLegend Cat# 372706, RRID:AB_2632732, BioLegend Cat# 372710, RRID:AB_2632925), NKp46 (BioLegend Cat# 331916, RRID:AB_2561621, BioLegend Cat# 331930, RRID:AB_2566117), CD16 (BioLegend Cat# 302038, RRID:AB_2561578, BD Biosciences Cat# 555407, RRID:AB_395807), DNAM-1 (BioLegend Cat# 338316, RRID:AB_2616645), NKG2D (BioLegend Cat# 320816, RRID:AB_2562747), NKG2A (BioLegend Cat# 375120, RRID:AB_2888868), PD-1 (BioLegend Cat# 329938, RRID:AB_2563596), TIM-3 (BioLegend Cat# 345026, RRID:AB_2565717), LAG-3 (BioLegend Cat# 369314, RRID:AB_2629797), CD45 (BioLegend Cat# 368532, RRID:AB_2715892), GD2 (BioLegend Cat# 357304, RRID:AB_2561885), CD155 (BioLegend Cat# 337618, RRID:AB_2565815), CD112 (BioLegend Cat# 337414, RRID:AB_2565732), CD33 (BD Biosciences Cat# 555450, RRID:AB_395843), CD83 (BD Biosciences Cat# 556910, RRID:AB_396534), HLA-DR (BD Biosciences Cat# 347364, RRID:AB_400292), CD14 (BioLegend Cat# 367144, RRID:AB_2810580), CD163 (BioLegend Cat# 333622, RRID:AB_2563612), CD20 (BD Biosciences Cat# 555622, RRID:AB_395988), CD107a (BioLegend Cat# 328630, RRID:AB_2562109), Fas-L (BioLegend Cat# 306412, RRID:AB_2716105), IFN-γ (BioLegend Cat# 506538, RRID:AB_2801098), p-mTOR (Ser2448) (Thermo Fisher Scientific Cat# 25–9718-42, RRID:AB_2573550), anti-CD247 (pY142) (BD Biosciences Cat# 558486, RRID:AB_647104), Annexin V (BioLegend Cat# 640950), 7AAD (BD Biosciences Cat# 559925, RRID:AB_2869266), FVS620 (BD Biosciences Cat# 564996, RRID:AB_2869636), FVS780 (BD Biosciences Cat# 565388, RRID:AB_2869673). The GD2.CAR was detected using Alexa Fluor647-conjugated 1A7 idiotypic antibody and FRP5 antibody (17,18), or 1A7 idiotypic antibody followed by Rat-Anti Mouse IgG1 (BD Biosciences Cat# 340270, RRID:AB_400010). Cells were stained with antibody for 15 minutes at 4 °C. Samples were fixed using BD Cytofix™ Fixation Buffer. All samples were acquired on a Gallios™ Flow Cytometer (Beckman Coulter) or FACSCanto II (BD Biosciences) and data was analyzed using FlowJo Analysis Software v11 (BD Biosciences, RRID:SCR_008520).

Intracellular staining.

NK cells were fixed (BD Phosflow™ Fix Buffer I, BD Biosciences Cat# 557870, RRID:AB_2869102), washed (BD Phosflow™ Perm/Wash I, (BD Biosciences Cat# 557885, RRID:AB_2869104), permeabilized with pre-chilled 100% methanol (BD Phosflow™ Perm Buffer II, (BD Biosciences Cat# 558052, RRID:AB_2869119) for 20 minutes on ice, and then washed twice. For phospho-FACS, cells were stained with either p-mTOR (Ser2448) or anti-CD247 (pY142) for 30 min at room temperature in the dark. For intracellular cytokine staining, NK cells were fixed, washed and permeabilized (eBioscience™ Foxp3/ Transcription Factor Staining Buffer Set, eBioscience™ Cat# 00–5523-00) and stained with anti-IFN-γ for 30 mins in the dark at 4 °C. For viability, cells were stained with either 7-AAD, FVS620, or FVS780.

Expansion of human NK cells.

Whole peripheral blood mononuclear cells (PBMCs) or CD3-depleted PBMCs (Human CD3 Positive Selection Kit II, Stem Cell Technologies Cat# 17851) obtained from 10 healthy blood donors consented under Baylor College of Medicine IRB-approved protocols, were co-cultured with irradiated (100 Gy) K562-mb15–41BB-L in a 1:10 (NK cell:irradiated tumor cell) ratio in G-Rex® cell culture devices (Wilson Wolf, St. Paul, MN, USA Cat# 80040S) for 6 days in Stem Cell Growth Medium (CellGenix Cat# 20802–0500) supplemented with 10% FBS and 500 IU/ml IL-2. For some experiments that did not require further genetic engineering, NK cells were expanded for 7 days with either K562-mb15–41BB-L or K562–41BB-CS or preactivated with IL-12 (10 ng/ml), IL-15 (50 ng/ml), and IL-18 (50 ng/ml) for 16 hours followed by IL-15 (10 ng/ml) for 6 days.

gRNA Design.

gRNAs targeting TIGIT were designed using the webtool (https://www.crisprscan.org/). Both TIGIT gRNAs utilized in the study target exon 2 (5’-TAATACGACTCACTATAGGTGACCGTGAACGATACAGGTTTTAGAGCTAGAA-3’) and (5’-TAATACGACTCACTATAGGGGGCCACTCGATCCTTGAGTTTTAGAGCTAGAA-3’). The DNAM-1 gRNA (5’-TAATACGACTCACTATAGGTCTTCTTCATGTATACAGGTTTTAGAGCTAGAA-3’) was based on a design previously described (19). Conditions containing Cas9 without any scrambled gRNA were used as a negative control.

CRISPR-mediated gene deletion and retroviral transduction of human expanded NK cells.

Cell Suspensions on day 6 were harvested and 3–4 × 10^6 NK cells were electroporated with Alt-R® S.p. Cas9 Nuclease V3 (2ug) (IDT Cat# 1081059), gRNAs (1ug/gRNA) and 1ul of Alt-R® Cas9 Electroporation Enhancer (IDT Cat# 10007805) in an electroporation strip (P3 Primary Cell 4D-Nucleofector™ X Kit S Cat# V4XP-3032) using the program DN-100 on the Lonza 4D-Nucleofector system. The protocol was adapted from a previously described paper (20). Immediately after electroporation, NK cells were transduced with SFG-based retroviral vectors, as previously described (13) for 4 days in Stem Cell Growth Medium. TIGIT^{KO} efficiency was validated using FACSCanto II (BD Biosciences) and ImageStream analysis by ImageJ v1.54k . Data was analyzed using FlowJo Analysis Software v11 (BD Biosciences). For *in vivo* experiments, the transduced cell population was subjected to secondary expansion to generate adequate cell numbers in G-Rex® devices at a 1:1 NK cell:irradiated K562-mb15–41BB-L ratio with 100 IU/ml IL-2 for 7 days in complete-Stem Cell Growth Medium.

Induction of human MDSCs and M2-TAMs *ex vivo*.

CD14+ PBMCs autologous to CD3-depleted PBMCs used for NK cell expansions were first isolated from healthy blood donors using magnetic column selection (Miltenyi Biotec) and incubated on 24-well tissue culture plates for 7 days with a GM-CSF (20 ng/mL) and IL-6 (20 ng/mL) cytokine cocktail known to induce MDSCs, as previously described (21). M2 macrophages were generated from a similar starting CD14+ PBMC material using M-CSF (100 ng/ml) and IL-4 (20 ng/ml).

Degranulation and IFN-γ assay.

NK cells were co-cultured with GFP-tagged LA-N-1 neuroblastoma and *ex vivo* generated MDSCs in a 1:4:8 NK cell:MDSC:Tumor ratio in a 24-well plate for 24 hours at 37 °C. Anti-CD107a added directly to the tri-culture on hour 20. The protein transport inhibitor Brefeldin A (Biolegend Cat# 420601) was added on hour 21 to the tri-culture and incubated for 3 hours at 37 °C. Intracellular IFN-γ and CD107a were analyzed using flow cytometry after the conclusion of the assay.

Tumor cytotoxicity kinetics assay.

Unmodified, TIGIT^{KO}, GD2.CAR-, and TIGIT^{KO}GD2.CAR-NKs labeled with a fluorescent dye (CellTracker™ Red CMTPX Dye, Invitrogen Cat# C34552) were incubated with green fluorescent protein (GFP)-CHLA255 neuroblastoma or GFP-H1650 lung carcinoma at 37 °C in either a 4:1 or 1:4, and 1:20 NK cell:tumor ratio respectively for 96 hours on a 24-well tissue culture plate. Kinetic live-cell imaging cytotoxicity assays were conducted on an IncuCyte S3 Live-Cell Analysis System (Sartorius). Tumor cell growth was determined by measuring green object area per well.

Generation of an M2-TAM containing TiME culture.

PBMC-derived CD14+ monocytes, and GFP-expressing CHLA255 neuroblastoma or H1650 lung carcinoma were incubated at a 1:1 monocyte:tumor ratio in a 24-well tissue culture plate for 72 hours at 37 °C with 100 ng/ml M-CSF to promote monocyte survival. After 72 hours, a monolayer containing CHLA255 or H1650 admixed with tumor-induced M2-TAMs was established and TiME properties such as inhibitory cytokine mileu, M2-TAM phenotype, and lymphocyte suppression were confirmed by multiplex ELISA of culture supernatants, M2-TAM flow cytometry, and IncuCyte assay, respectively. After 72 hours, half of the media in each well was replaced with fresh complete-RPMI to maintain viability of TiME in the cultures. Kinetic live-cell imaging cytotoxicity assays were conducted on an IncuCyte S3 Live-Cell Analysis System (Sartorius). Tumor cell growth was determined by measuring green object area per well.

TiME Tumor Cytotoxicity Assay.

After TiME establishment described above, 1×10^5 of fluorescently dye labeled (CellTracker™ Red CMTPX Dye, Invitrogen) Unmodified, TIGIT^{KO}, GD2.CAR-, and TIGIT^{KO}GD2.CAR-, DNAM-1^{KO}GD2.CAR-, and TIGIT^{KO}DNAM-1^{KO}GD2.CAR-NKs were added to each well with 10 ng/ml IL-15 and incubated at 37 °C for 96 hours on the 24-well tissue culture plate. Kinetic live-cell imaging cytotoxicity assays were conducted on an IncuCyte S3 Live-Cell Analysis System (Sartorius). In some experiments, all NK cells were incubated for 72 hours at 37 °C prior to being transferred to a new freshly prepared TiME. This step was repeated one more time. After every transfer, 10 ng/ml IL-15 was provided to maintain NK survival. The green object area reading when NK cells are added to a prepared TiME (0 hour) is 'Green Object Area (T0)'. The green object area reading after 72 hours of NK-TiME co-culture is 'Green Object Area (T72)'. Therefore, the fold change in tumor burden is calculated as follows:

$$Fold\ \ Change(Tumor\ \ Burden) = \frac{Green\ \ Object\ \ Area\ \ (T72)}{Green\ \ Object\ \ Area\ \ (T0)}$$

In vivo tumor immune microenvironment xenograft model.

6–8-week-old male and female NSG-SGM3 mice (The Jackson Laboratory, Strain # 013062, RRID:IMSR_JAX:013062) were implanted subcutaneously in the dorsal right flank with 1×10^6 LA-N-1 neuroblastoma cells admixed with 5×10^5 CD14+ PBMCs, suspended in basement membrane Matrigel (Corning). LA-N-1 neuroblastoma cells do not express ligands for the most highly expressed NK activating receptor, NKG2D, but do express GD2 and both TIGIT ligands at comparable levels to CHLA255 neuroblastoma (but which also has high levels of NKG2D ligands). This allows control for other NK activating receptors while isolating the effect of either TIGIT presence or absence on CARmediated effects. Matrigel basement membrane was important in keeping tumor and CD14+ cells confined to establish a localized solid TiME. This TME model of an established tumor mass was used as it better mimics challenges seen in patients with advanced solid tumors compared to traditional tumor xenografts (18). 12–14 days later, when tumors measured ~100 mm^3 by caliper measurement, mice were divided into 5 groups (n = 5–6 mice/group) and injected intratumorally with 3 doses of 8×10^6 unmodified, $TIGIT^{KO}$, GD2.CAR-, and $TIGIT^{KO}$GD2.CAR-NK or no NK cells. Each dose was injected every 5 days. Tumor growth was measured 2–3 times weekly by caliper measurements. IL-15 was also injected intraperitoneally 2–3 times weekly (1 μg/mouse) to enable NK cell survival. In experiments examining the ability of $TIGIT^{KO}$GD2.CAR-NKs to persist in the TiME, tumors were harvested from mice (n = 3/group) 4 days after the first NK cell dose. Tumors were digested, counted with a hemocytometer, and stained for GD2, CD45, CD16, CD3, 1A7, and TIGIT. Data was acquired via flow cytometry with absolute cell counts measured by multiplying the frequencies of various cell populations by the total cell count enumerated prior to flow cytometry. The animal protocol was approved by Baylor College of Medicine Institutional Animal Care and Use Committee and mice were treated in strict accordance with the institutional guidelines for animal care.

Annexin V assay.

Unmodified, $TIGIT^{KO}$, GD2.CAR-, and $TIGIT^{KO}$GD2.CAR-NK were co-cultured in a prepared CHLA255 TiME for 72 hours. After incubation, NK cells were stained with DNAM-1, 1A7, and TIGIT prior to being resuspended in Annexin V binding buffer (BD Biosciences) and subsequent staining with Annexin V (BioLegend) for 15 min in the dark at room temperature. Data was acquired on FACSCanto II (BD Biosciences) and analyzed with FlowJo software v11. To determine viability, NK cells were stained with 7AAD just prior to sample acquisition on the flow cytometer.

Caspase3/7 assay.

Unmodified, $TIGIT^{KO}$, GD2.CAR-, and $TIGIT^{KO}$ GD2.CAR-NK were co-cultured in a prepared CHLA255 TiME for 72 hours. After incubation, NK cells were stained with NKp46, 1A7, and TIGIT prior to being staining with Caspase3/7 (CellEvent® Caspase-3/7 Green Flow Cytometry Assay Kit, Invitrogen Cat# C10740) for 25 min at 37 °C followed by

SYTOX® AADvanced™ (Invtrogen Cat# S10349) for 5min at 37 °C to assess viability. Data was acquired on FACSCanto II (BD Biosciences) and analyzed with FlowJo software v11.

pMTOR assay.

Unmodified, TIGIT^{KO}, GD2.CAR-, and TIGIT^{KO}GD2.CAR-NK were co-cultured in a prepared CHLA255 TiME for 72 hours. After incubation, NK cells were stained with NKp46, 1A7, and TIGIT prior to being staining fixation and permeabilization followed by pMTOR staining as detailed above. Data was acquired on FACSCanto II (BD Biosciences) and analyzed with FlowJo software v11.

Mitochondrial mass assay.

Unmodified, TIGIT^{KO}, GD2.CAR-, and TIGIT^{KO}GD2.CAR-NK were co-cultured in a prepared CHLA255 TiME for 72 hours. After incubation alone or with the TiME, NK cells were stained with CD16, 1A7, and TIGIT prior to being staining with a Mitochondrial labeling dye (MitoTracker™ Green FM Dye, Invitrogen Cat# M7514) for 30 min at 37 °C. Data was acquired on FACSCanto II (BD Biosciences) and analyzed with FlowJo software.

Cell Avidity Assay.

CHLA255 cells were seeded into a poly-l-lysine (Sigma-Aldrich) coated piezo chip from LUMICKS. CHLA255 cells adhered for 2 hours. Unmodified, TIGIT^{KO}, GD2.CAR-, and TIGIT^{KO} GD2.CAR-NK were labeled with CellTrace Far Red (Invitrogen, Cat# C34564) at a 1:1,000 dilution. The CHLA255 bound piezo-chip was loaded onto the z-MOVI single-cell avidity analyzer. CellTrace labeled NK cells were injected into the chip and allowed to incubate on the CHLA255 cells for 5 min. After this time, NK cells were subjected to increasing acoustic force ramp from 0 to 1,000 pN over 3 min. Individual NK cells were observed, and the force requirement for detachment was determined based on any individual NK cell leaving the focal plane, as previously described (22).

Nanostring Gene Expression Analysis.

Unmodified, TIGIT^{KO}, GD2.CAR-, and TIGIT^{KO} GD2.CAR-NK were co-cultured with CHLA255. RNA was extracted from NK cells prior to tumor exposure and after 72 hours tumor exposure and CHLA255 depletion (EasySep™ PE Positive Selection Kit II, Stem Cell Technologies, Cat# 17684). RNA eluates were stored at −20°C until further processing. When ready, samples were analyzed at the Baylor College of Medicine Genomic and RNA Profiling Core on the Nanostring genomics platform. Results were analyzed using the nCounter Advanced Analysis in conjunction using nSolver Analysis Software (RRID:SCR_003420).

TIMING Assay:

Nanowell arrays on a chip were fabricated as previously reported (23). The chip was placed in a plasma chamber (Harrick Plasma Inc; Ithaca, NY) for two min and treated with PLL[20]-g[3.5]- PEG(2)/PEG(3.4)- biotin(50%) (SuSoS; Zurich, Switzerland). Target and effector cells were washed three times with PBS prior to staining. We labeled the target (CHLA255 tumor) and effector (CAR-NK) cells with PKH26 Red (Sigma-Aldrich, Cat#

PKH26GL-1KT) and PKH67 Green (Sigma-Aldrich, Cat# PKH67GL-1KT), respectively. Cells were then washed three times in full media and resuspended at a density of 1 million cells/mL. Effector and target cells were loaded onto the nanowell arrays and resuspended in IMDM, 10% FBS and Annexin V Alexa Fluor 647. No cytokines were added. The chip was imaged in BrightField, Alexa Fluor 488, TexasRed, and Cy5 using a Zeiss Axio Observer (Oberkochen, Germany) for 10 hrs over 99 time points. Image analysis and cell tracking were performed as previously reported (24).

Statistical Analysis.

All numerical data are represented as mean ± standard error of the mean (SEM) of either number of donors or experimental replicates, as indicated. Unpaired two-tailed t-test, paired two-tailed t-test, one-way or two-way ANOVA with Tukey's multiple comparison test, log-rank (Mantel-Cox) test, and chi-square test as indicated, were used to determine significance of differences between means with $p < 0.05$ representing a significant difference. All statistical analyses were performed with GraphPad Prism version 10.4.2 (RRID:SCR_002798).

Data Availability.

The data generated in this study are available upon request from the corresponding author. Transcriptomic data on various human NK cell populations before and after exposure to TiME conditions was uploaded to Gene Expression Omnibus (GEO) (RRID:SCR_005012) (Accession Number: GSE288177).

RESULTS

TIGIT is persistently upregulated upon NK cell activation during *ex vivo* CAR-NK manufacture.

To assess the effect of NK cell activation and expansion requirements for CAR-NK manufacture on TIGIT expression, we co-cultured human peripheral blood NK cells with the K562-mb15–41BB-L feeder cell line in the presence of 500 IU/mL IL-2 for 7 days and assessed TIGIT expression by flow cytometry. TIGIT was upregulated on activated/expanded NK cells in each donor examined in both percentage of NK cells in the population that expressed TIGIT and the level of surface TIGIT per NK cell (Fig. 1A). TIGIT expression has historically been viewed to be activation dependent, but this data is limited to endogenous NK cells. To determine the fate of TIGIT expression on *ex vivo* activated/expanded NK cells, we examined TIGIT expression over time on activated NK cells. NK cells that received no further activation for 7 days continued to express elevated levels of TIGIT (Fig. 1B). NK cells that received additional activating signals via 100 IU/mL IL-2 similarly expressed elevated levels of TIGIT but to no higher levels than NK cells without additional stimulation, suggesting persistence of TIGIT expression on NK cells expanded for CAR-NK manufacture. To determine if TIGIT upregulation in NK cells was a result of our specific NK activation/expansion protocol, we assessed NK cell TIGIT expression using two other published protocols commonly used to activate and expand NK cells: K562-CD80-CD86–41BB-L feeder cells plus recombinant IL-21 (25), and the recombinant cytokine cocktail of IL-12+IL-15+IL-18 known to induce memory-like NK cells for adoptive transfer

(26). The NK cell expansion rate mediated by each expansion protocol was consistent with previous reports (Fig. 1C), reflecting adequate activation/expansion. While not statistically significant for the other published methods, TIGIT expression was upregulated in all three expansion systems (Fig. 1D), suggesting that TIGIT upregulation is seen consistently during *ex vivo* activation/expansion irrespective of the type of NK manufacturing protocol used. Having characterized TIGIT upregulation on NK cells developed for therapeutic purposes, we next wanted to profile the expression of the most common TIGIT ligands, CD155 and CD112, in patients with GD2+ solid tumors. We assessed inhibitory myeloid cells as well as tumor cells from freshly resected pediatric neuroblastoma tissue obtained at first resection for CD155 and CD112 expression. MDSCs and M2-TAMs were found to highly express both CD155 and CD112, reproducibly at levels higher than tumor cells themselves (Fig. 1E). Our combined data suggest that the TIGIT-TIGIT ligand axis has the potential to inhibit the function of adoptively transferred therapeutic NK cells in solid tumors.

Genetic deletion of TIGIT does not alter CAR-NK expansion or phenotype.

Given persistent elevation in TIGIT in NK cells undergoing activation/expansion protocols for CAR-NK manufacture and the high expression of TIGIT ligands in solid TiMEs, we hypothesized that deletion of TIGIT would improve CAR-NK efficacy by rendering CAR-NKs resistant to inhibitory signaling. We developed a sequential CRISPR/Cas9 and retroviral transduction protocol to delete TIGIT in human CAR-NKs. Primary human peripheral blood NK cells were expanded with a K562-mb15–41BB-L feeder cell line and IL-2 for 6 days. Using electroporation, we delivered a payload of Cas9 protein and two gRNAs targeting exon 2 of *TIGIT* in expanded NK cells. This was followed immediately by retroviral transduction of NK cells with a published GD2-targeting CAR (13) (Fig. 2A). Using this approach, we consistently achieved >90% TIGIT knockout efficiency (avg. 84% TIGIT+ cells in GD2.CAR-NK vs 6.5% TIGIT+ cells in TIGIT^{KO}GD2.CAR-NK) and >60% GD2.CAR transduction efficiency (avg. 60.8% in GD2.CAR-NKs vs 63.4% TIGIT^{KO}GD2.CAR-NKs) (Fig. 2B-C, Supplementary Fig. S1B-C). To determine if TIGIT deletion impacted proliferation during manufacture, we expanded TIGIT^{KO}CAR-NKs with K562-mb15–41BB-L and IL-2. No significant differences were seen in fold-expansion of TIGIT^{KO}GD2.CAR-NKs compared to GD2.CAR-NKs (avg. 83-fold expansion TIGIT^{KO}GD2.CAR-NK vs. 73-fold expansion GD2.CAR-NK) after 7 days of expansion with K562-mb15–41BB-L feeder cell line in the presence of 100 IU/mL (Fig. 2D). NK cells express multiple activating and inhibitory germline encoded receptors that hold key functions in enabling NK cell cytotoxicity. We assessed if TIGIT deletion in generated GD2.CAR-NKs altered their expression of activating and inhibitory receptor repertoires. TIGIT deletion did not alter surface expression of the activating receptors DNAM-1, NKG2D, and the inhibitory receptors NKG2A, TIM-3, PD-1 or LAG-3 on TIGIT^{KO}GD2.CAR-NKs after *ex vivo* manufacture (Fig. 2E).

TIGIT deletion in GD2.CAR-NKs improves anti-tumor activity.

To evaluate their anti-tumor activity, TIGIT^{KO}GD2.CAR-NKs were co-cultured with GFP-expressing CHLA255 neuroblastoma for 96 hours in an NK cell-favoring ratio (E:T of 4:1) and a tumor-favoring ratio (E:T of 1:4). While all NK cell conditions successfully eliminated tumor within 24 hours at NK cell-favoring ratios, only TIGIT^{KO}GD2.CAR-NKs eliminated tumor at tumor-favoring ratios and exhibited superior tumor-killing kinetics, regressing tumor within 12 hours compared to no tumor regression by GD2.CAR-NKs (Fig. 3A). To assess if upregulation of activating receptors, downregulation of inhibitory receptors, or changes in CAR expression contributed to the enhanced tumor killing by TIGIT^{KO}GD2.CAR-NKs, we assessed NK cell receptor expression before and after tumor co-culture. After 96 hours of tumor exposure, no differences were observed in GD2 CAR expression between TIGIT expressing- or TIGIT^{KO}GD2.CAR-NKs (Fig. 3B). Additionally, no significant changes were observed in the expression of DNAM-1, NKG2D, NKG2A, TIM-3, PD-1, or LAG-3 on TIGIT^{KO}GD2.CAR-NKs after tumor exposure when compared to their expression on GD2.CAR-NKs (Fig. 3C). TIGIT deletion did not result in increased degranulation (CD107a), IFN-γ expression or Fas-L expression on GD2.CAR-NK after long-term culture (Fig. 3D-F). Overall, TIGIT deletion did not result in any significant compensatory changes in NK receptors or receptor functions that would account for the observed enhanced CAR-NK tumor control.

TIGIT^{KO}GD2.CAR-NKs limited tumor growth and expanded in the presence of immunosuppressive myeloid cells.

To evaluate the effect of TIGIT deletion on the anti-tumor activity of CAR-NKs in a microenvironment that mimics TiME immunosuppression and the increased TIGIT ligand expression seen in patient tumors, we co-cultured CAR-NKs with the TIGIT ligand-expressing NK cell-insensitive neuroblastoma line, LA-N-1, in combination with TIGIT ligand-expressing monocytederived MDSCs for 24 hours with an effector:tumor:myeloid ratio of 1:2:2 (Supplementary Fig. S2A-C). The inclusion of MDSCs in tumor co-cultures led to decreased CD107a and IFN-γ expression by GD2.CAR-NKs. In contrast, TIGIT^{KO}GD2.CAR-NKs maintained IFN-γ expression. However, TIGIT deletion did not reverse inhibition of GD2.CAR-NK degranulation. In the presence of tumor and MDSCs, GD2.CAR-NKs exhibited decreased CD107a degranulation, and IFN-γ production as compared to in the presence of tumor alone, reflecting MDSC-based immunosuppression (Fig. 4A). In contrast, TIGIT^{KO}GD2.CAR-NKs maintained their ability to secrete IFN-γ in the presence of tumor and MDSCs. However, no differences were observed in percentage of NK cells degranulating in the presence to tumor and MDSCs between TIGIT-expressing and TIGIT deleted CAR-NKs. To more directly assess whether TIGIT deletion would enhance CAR-NK function in harsh tumor microenvironments, we developed a TiME co-culture assay that mimicked the immunosuppressive environment of solid tumors with suppressive myeloid infiltrates (Supplementary Fig. S3A). We co-cultured CD14+ monocytes for three days with a CHLA255 neuroblastoma cell line previously demonstrated to secrete high amounts of TGF-β1 and IL-6, both of which have been implicated in tumor-induced polarization of intratumor myeloid cells (27–29). After three days of co-culture, the monocytes were skewed to an inhibitory M2-TAM-like phenotype (Supplementary Fig. S3B-D). The inclusion of tumor-influenced M2-TAMs produced a highly immunosuppressive environment that resulted in decreased CARNK tumor cytotoxicity compared to that of CAR-NKs in tumor cultures without M2-TAMs (Supplementary Fig. S3E-F). To this established TiME co-culture, we added TIGIT^{KO}GD2.CAR-NKs (manufactured from same donor as the macrophages in the TiME co-culture to avoid alloreaction) at a tumor-favoring E:T ratio of 1:10. Unmodified NK cells, GD2.CAR-NKs, and TIGIT^{KO}NK cells without a CAR were used as controls. TIGIT^{KO}GD2.CAR-NKs displayed robust tumor killing, enhanced killing kinetics, and both early and rapid expansion compared to controls in this TiME culture system (Fig. 4BC). To improve the rigor of the study, the enhanced anti-tumor kinetics conferred by TIGIT deletion in the TiME was also validated in HER2.CARNKs against the HER2+ lung cancer cell line H1650 (Supplementary Fig.S4A). As in previous assays, these enhancements were not due to increased expression of the CAR or changes in expression of NK cell germline receptors (Fig. 4D-E). We next evaluated the ability of TIGIT^{KO}GD2.CAR-NKs to control tumor after repeated TiME exposure, assessing their ability to mediate effective serial killing. TIGIT^{KO}GD2.CAR-NKs were co-cultured over three sequential 72-hour TiME co-cultures. Consistent with our previous data, TIGIT^{KO}GD2.CAR-NKs exhibited improved tumor killing in the first TiME co-culture compared to GD2.CAR-NKs (Fig. 4F, TiME #1). While all control conditions failed to control tumor in successive TiME co-cultures, TIGIT^{KO}GD2.CAR-NKs exhibited the best tumor control (Fig. 4F, TiME #2) but eventually faltered with successive challenge (Fig. 4F, TiME #3).

TIGIT^{KO}GD2.CAR-NKs control tumor and prolong survival in mice bearing TiME xenografts.

We injected NSG-SGM3 mice subcutaneously with LA-N-1 tumor admixed with CD14+ monocytes in Matrigel extracellular matrix solution. When tumors reached ~100mm^3, tumors formed a suppressive TiME-like matrix rich in suppressive stroma, inhibitory myeloid cells, and a rich suppressive cytokine milieu (21). To assess the anti-tumor activity of TIGIT^{KO}GD2.CAR-NKs directly within a solid TiME without the confounding effects of immune cell trafficking, mice were treated with three intratumor injections of NK cells every 5 days (see Fig. 5A for treatment schema). While immunosuppressive TiME xenograft-bearing mice treated with control NK cells exhibited tumor growth and early death, TIGIT^{KO}GD2.CAR-NK-treated mice experienced modest control of their tumor burden, with 2 of 5 mice experiencing durable tumor control (Fig. 5B) and extended survival (Fig. 5C). To determine if differences in GD2.CAR expression or NK persistence could explain the observed differences in response between the groups of treated mice, tumors were harvested 4 days after the first NK cell injection and analyzed by flow cytometry. While we observed no differences in GD2 CAR expression on intratumor NK cells among the CAR-containing groups (32.8% on GD2.CAR-NKs and 37.9% on TIGIT^{KO}GD2.CAR-NKs; Fig. 5D), there were higher numbers of intratumor CAR-NKs in mice treated with TIGIT^{KO}GD2.CAR-NKs compared to those treated with GD2.CAR-NKs (Fig. 5E).

TIGIT^{KO}GD2.CAR-NKs downregulate genes associated with cell adhesion and exhibit decreased CAR-mediated synapses with tumor.

TIGIT deletion in CAR-NKs enhances CAR-NK expansion and ability to control tumor in harsh TiMEs despite similar levels of degranulation, CAR molecule expression, and Fas-L surface expression. We also observed comparable levels of Caspase3/7, Annexin V, mitochondrial mass, and pMTOR levels (Supplementary Fig. S4B-E). Further, the contribution of the receptor DNAM1, an activating receptor that competes with TIGIT for its ligands, was inconsequential to CAR-NK antitumor activity upon TIGIT deletion in NK cells (Supplementary Fig. S4F). To determine how TIGIT improved CAR-NK activity in the TME, we performed bulk global gene expression analysis. We isolated total RNA from unmodified, $TIGIT^{KO}$, GD2.CAR-NK, or $TIGIT^{KO}$GD2.CAR-NKs before and after 72 hours of co-culture with neuroblastoma tumor cells. Principal component analysis revealed that while tumor exposure impacted gene expression, TIGIT deletion did not significantly alter gene expression patterns in either non-CAR or GD2.CAR-expressing NK cells (Fig. 6A). When comparing gene expression between $TIGIT^{KO}$GD2.CAR-NKs and GD2.CAR-NKs after tumor exposure, significant differences were noted in both upregulated and downregulated genes. Global gene score analysis showed decreased expression of genes related to cell adhesion, apoptosis, and oxidative stress response in $TIGIT^{KO}$GD2.CAR-NKs (Fig. 6B). Consistent with the robust downregulation in cell adhesion score, the cell adhesion and immunological synapse protein VCAM1, was the most downregulated gene in $TIGIT^{KO}$GD2.CAR-NKs (Fig. 6C, Supplementary Fig. S4G-H). To verify that this downregulation resulted in reduced % of VCAM-1 surface protein expression, we co-cultured $TIGIT^{KO}$GD2.CAR-NKs or control NKs with CHLA255 for 72 hours. We observed no significant downregulation of VCAM-1 protein in non-CAR-expressing NKs, irrespective of TIGIT expression. In contrast, compared to TIGIT-expressing GD2.CAR-NKs, $TIGIT^{KO}$GD2.CAR-NKs exhibited lower levels of surface VCAM-1 protein (Fig. 6D, Supplementary Fig. S4I). An assessment of its expression level per NK cell revealed a modest decrease in surface VCAM-1 on both non-CAR and CAR-NK cells after CHLA255 exposure for 72 hours.

Given these data suggesting differences in NK adhesion to targets, we wanted to directly test the effect of TIGIT deletion on immune synapse strength. We first evaluated NK cell structural avidity at the single-cell level using a LUMICKS z-MOVI cell avidity analyzer, which measures the binding force between effector and target cells, to determine if TIGIT deletion alters CAR-NK interactions with tumors via synapse formation or release. NK cells lacking TIGIT (either non-CAR or CAR-expressing) exhibited lowered binding capacities to CHLA255 compared to their TIGIT-expressing counterparts (Fig. 7A). We further interrogated the effect of TIGIT deletion on synapse stability by using the Time-lapse Imaging Microscopy In Nanowell Grids (TIMING) assay (23,24). We utilized this assay to evaluate the duration of GD2.CAR-NK cell mediated synapses against CHLA255 neuroblastoma, and the resulting time taken to mediate tumor death (Fig. 7B). We assessed the time of contact between NK cells and tumor targets ($T_{Contact}$), and the time taken by NK cells by induce tumor cell death (T_{Death}) in nanowells at an E:T ratio of 1:1 (Fig. 7C). When comparing NK cells without CARs, individual $TIGIT^{KO}$ NK cells (green) exhibited reduced duration of contact with the tumor cells (138 ± 12 minutes, mean ± SEM) in comparison to unmodified NK cells (198 ± 12 minutes) (Fig. 7C). Similarly, individual $TIGIT^{KO}$ NK cells also killed tumor cells faster (120 ± 18 minutes) in comparison to unmodified NK cells (192 ± 24 minutes) (Fig. 7C). We next compared GD2.CAR-expressing NK cells using the TIMING assays. Consistent with our results with non-CAR NK cells, $TIGIT^{KO}$ GD2.CAR-NK cells exhibited the shorter contact durations (138 ± 18 minutes, mean ± SEM) and faster killing of tumor targets (102 ± 18 minutes) compared to control GD2.CAR-NK cells which exhibited a longer contact time (228 ± 24 minutes) with a slower kill time (138 ± 18 minutes) (Fig. 7C).

To determine if decreased contact duration may allow for more rapid engagement of multiple tumor targets, TIGIT^{KO}GD.CAR-NKs were cultured in nanowells at E:T ratios of 1:2. In contrast to control GD2.CAR-NKs, TIGIT^{KO}GD2.CAR-NKs consistently formed successive shorter-lived functional immunological synapses with CHLA255 tumor targets (Fig. 7D-E). Consistent reductions in synapse duration between TIGIT^{KO}GD2.CAR-NKs and tumor targets were only observed in contact 1 while contact 2 exhibited a more diverse pattern of synapse durations. GD2.CAR-NK cells exhibited a longer cumulative time spent to contact and disengage from both tumor targets compared to TIGIT^{KO}GD2.CAR-NK cells (240 ± 42 minutes for GD2.CAR-NK vs 96 ± 36 minutes for TIGIT^{KO}GD2.CAR-NK). The swift termination of the TIGIT^{KO}GD2.CAR-NK immunological synapse did not come at the expense of its tumor cytotoxicity. Instead, we observed that a greater proportion of TIGIT^{KO}GD2.CARNK cells eliminated 2 tumor targets when compared to their TIGIT-expressing controls (87% TIGIT^{KO}GD2.CAR-NK vs 71% GD2.CAR-NK) (Fig. 7F). Collectively, these results from single cell assays demonstrate that knocking out TIGIT in NK cells (with or without CAR) enables formation of efficient synapses with tumor cells resulting in fast killing and serial killing.

DISCUSSION

In this report, we show that human CAR-NKs from which TIGIT was genetically deleted evade TIGIT-mediated suppression found in solid tumor microenvironments. We demonstrate that TIGIT^{KO}GD2.CAR-NKs exhibit both enhanced tumor control and proliferation in solid TiMEs that typically inhibit TIGIT-expressing CAR-NKs both *in vitro* and in xenograft-bearing mice. Further, we present data that reveal a previously unknown role for TIGIT in modulating cell adhesion of CAR-NKs to their tumor targets. GD2.CAR-NKs without TIGIT downregulated genes associated with cell adhesion and exhibited reduced cell avidity against tumor targets, conferring a more efficient serial killing capacity to CAR-NKs. Thus, our genetic engineering approach to interrogate TIGIT-mediated effects on CAR-NKs within solid TiMEs revealed a non-canonical mechanism by which TIGIT affects NK cells expressing CARs and enabled an approach that might improve efficacy of CAR-NKs targeting solid tumors.

Our study reveals different mechanisms than canonical TIGIT signaling by which TIGIT may impair CAR-mediated function in NK cells. Previous studies have used antibody blockade of CD155, CD112, or TIGIT in murine and human endogenous NK cells (30,31). These studies suggested that TIGIT inhibited NK cell anti-tumor activity by acting as a competitive receptor to the activating receptor DNAM-1 which also binds CD155 or CD112 on tumors, and thereby blocking DNAM-1-based activation of NK cells (32,33). In contrast to endogenous NK cells, our data suggest that DNAM-1 activation is not involved in enhancing NK cytotoxicity in the context of CAR-NKs, where the additional activating signals through the CAR molecule likely override any benefit from DNAM-1. Canonical TIGIT signaling also inhibits AKT-dependent survival via reduced apoptosis (34), activates the MTOR pathway, and upregulates genes associated with glycolysis (35). However, loss of TIGIT signaling in CAR-NKs did not increase phosphorylated MTOR nor result in apoptotic resistance or increased mitochondrial mass typically regulated by AKT(36). Taken together, our data suggests that the canonical inhibitory TIGIT signaling typically responsible for impairment of NK cytotoxicity is not responsible for CAR-NK inhibition and that an alternate role for TIGIT may exist specifically in the context of CAR expression.

Indeed, our data suggest that TIGIT may modulate CAR-NK adhesion to tumor targets. Previous work has shown that when endogenous NK cells engage targets via their natural receptors, TIGIT rapidly localizes to synapses rich in TIGIT ligands on target cells, thus preventing DNAM-1 from binding TIGIT ligands and activating NK cytotoxicity (37,38). In NK cells modified to express CARs, our data suggest that TIGIT localization to CAR-containing synapses does not serve to outcompete DNAM-1 but to enhance cell adhesion and cell avidity to tumor targets. Hence, the improved tumor control resulting from TIGIT deletion on CAR-NKs is not dependent on DNAM-1 activation and enhanced cytotoxic potential but rather on diminished VCAM-1 mediated cell adhesion and weakened cell avidity, which potentially allows for improved serial killing and more efficient tumor destruction. Mechanistically, the weakening of VCAM-1-based adhesion and avidity by loss of TIGIT at the immune synapse may occur due to decreased TIGIT-mediated inhibition of regulators of PI3 kinase activity resulting in downregulation of LFA-1, a critical initiator of NK cell synapse formation (39,40). However, it is unlikely that the impact of TIGIT deletion is solely limited to the regulation of adhesion molecules. The mere absence of TIGIT, a high affinity, synapse localized receptor may directly impact the avidity of CAR-NK cells with tumor targets by rendering unable to engage with CD155 and CD112 on the tumor. Our data showed that this weakened avidity was partially rescued after contact with subsequent tumor targets. This suggests that mechanisms directing the accelerated kinetics of the first CAR-NK-tumor synapses may be different from those that regulate subsequent synapses. This altered avidity might be influenced by the dynamic changes in the levels of intracellular ATP and serine/threonine kinases, as well as extracellular shuttling dynamics of CAR molecules which are critical to the maintenance of the immunological synapse (18,41). Additionally, under excess ligand availability, increased synaptic localization DNAM-1 may engage more efficiently to slightly enhance CAR-NK adhesion over multiple rounds of serial tumor killing. This highlights the utility of an alternate mechanism to enhance CAR-NK activity relative to previous approaches that attempted to improve CAR-NK cytotoxicity by strengthening immune synapse formation (22). Although increased cell avidity has typically been perceived as beneficial to CAR-mediated tumor killing, recent studies have shown that CAR-T cells possessing intermediate binding avidity compared to those with high avidity for their targets displayed the best tumor control owing to increased serial killing activity (42–45). Thus, the reduction in synapse strength exhibited by TIGIT^{KO}GD2.CAR-NKs allowing faster detachment times and more rapid serial killing may allow improved overall tumor control.

Loss of TIGIT expression also enhanced CAR-NK proliferation and persistence within the solid TiME. Persistence and proliferation were key factors for anti-tumor activity and objective responses in patients in two recent adoptive CAR-NK clinical trials against hematologic malignancies (4,5). In solid tumors, which contain TiMEs that would result in impaired proliferation of infiltrating NK cells, enhancement of CAR-NK proliferation is likely more critical to ensure adequate anti-tumor activity (46). We observed reduced CAR-NK proliferation in the presence of a TIGIT ligand-enriched TiME that was rescued by deletion of TIGIT. This can be explained by the fact that TIGIT typically suppresses the NF-kβ signaling responsible for CAR-induced NK proliferation (10,47). Indeed, this is supported by our data showing that TIGIT deletion improved proliferation and persistence of CAR-modified but not unmodified NK cells in the TiME. It is also possible that the improved persistence observed with TIGIT loss in CAR-NKs is due to decreased pro-apoptotic signaling. We observed that $TIGIT^{KO}$GD2.CAR-NKs after tumor exposure had decreased expression of the pro-apoptotic gene *BID* and *PDCD2*. This suggests that TIGIT may inhibit CAR-NK persistence by directly promoting *BID* transcription or by interfering with upstream negative regulators of BID including AKT, thus leading to impaired long-term persistence in the TiME (48,49). Loss of TIGIT can possibly reverse this process, analogous to the way that PD-1 blockade can improve persistence of CAR-T cells by reducing the upregulation of Fas receptor and pro-apoptotic genes induced after CAR-mediated activation (48,50–52).

In summary, we defined a unique role for TIGIT in genetically engineered human CAR-NKs. We have shown that generation of a *TIGIT*-deleted CAR-NK product is achievable and that $TIGIT^{KO}$GD2.CAR-NKs exhibit antitumor activity, expand, and persist within a TIGIT ligand-enriched suppressive solid tumor microenvironment. Our CAR-NK manufacturing protocol incorporating TIGIT deletion via CRISPR/Cas9 can be broadly applied to human NK cell products expressing any solid tumor CAR. Experiments investigating the anti-tumor effect of TIGIT deletion in numerous CAR antigen and tumor contexts *in vivo* are needed to more broadly apply our findings and are ongoing. We demonstrated that traditional inhibitory TIGIT signaling does not play a vital role in the context of CARs and does not account for the improved CAR activity observed in our study. Instead, we highlight a non-canonical role for TIGIT in modulating cell adhesion and immune synapse strength in CAR-NKs within TIGIT ligand-enriched TiMEs. These findings will hopefully guide new biology-driven strategies to overcome inhibitory NK receptors like TIGIT and improve efficacy of CAR-NKs against solid tumors.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Abbreviations used in this paper:

CAR	chimeric antigen receptor
PBMC	peripheral blood mononuclear cell
TiME	tumor immune microenvironment

E:T	effector-to-target ratio
MDSC	myeloid derived suppressor cell
M2-TAM	inhibitory tumor-associated macrophage
NK	natural killer
TIGIT	T-cell Immunoreceptor with Ig and ITIM
ITIM	immunoreceptor tyrosine-based inhibitory motif
ITT	immunoglobulin tail tyrosine

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Synopsis:

The authors demonstrate that TIGIT on CAR-expressing effector cells impairs antitumor functions via its effects on the tumor-effector cell synapse. Modulating inhibitory receptors like TIGIT in engineered NK cells that alter synapse strength may overcome TME mediated immunosuppression.

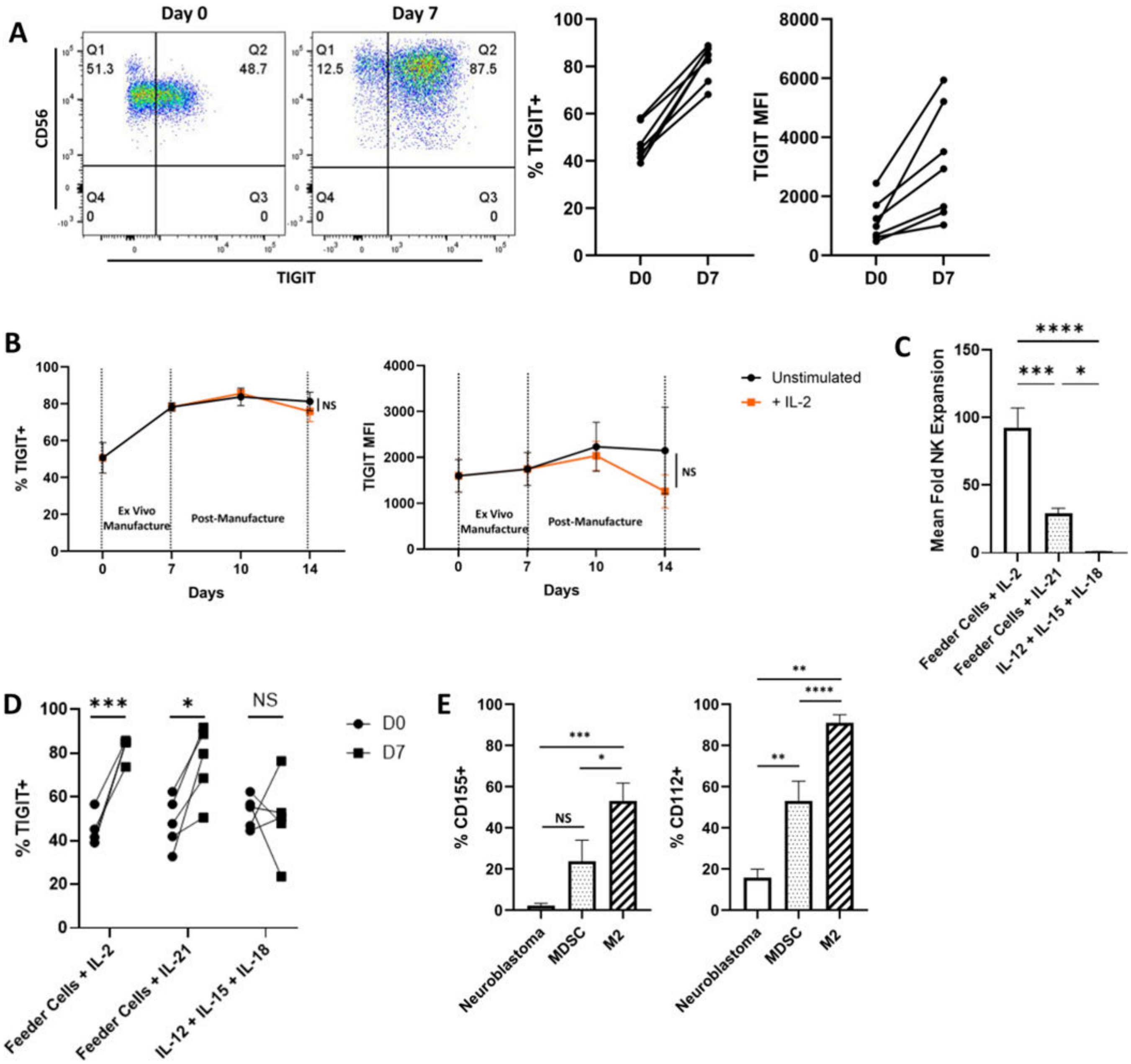

Figure 1. TIGIT is upregulated in human NK cells undergoing activation and expansion for CAR-NK manufacture.

(A) Representative flow cytometric dot plot from a single donor (left) and Mean change in TIGIT expression on PBMC-derived CD56+CD3- NK cells from n=7 healthy donors before (D0) and after 7 days (D7) of activation/expansion using the feeder cell line k562-mb15–41BB-L and IL-2,*$p<0.05$, ****$p<0.0001$. (B) Change in TIGIT expression on NK cells over time during and after ex vivo manufacture with and without IL-2 (n=5 healthy blood donors), NS, Non-significant. (C) Mean fold expansion of NK cells after 7 days of activation using k562-mb15–41BB-L and IL-2, k562–41BB-L and IL-21, or IL-12+IL-15+IL-18 cytokine cocktail (n=4–5 healthy donors), NS, Non-significant; *$p<0.05$, **$p<0.01$. (D) Flow cytometric analysis of TIGIT expression before and after 7 days of activation using k562-mb15–41BB-L and IL-2, k562–41BB-L and IL-21, or IL-12+IL-15+IL-18 cytokine cocktail (n=4–5 healthy donors), NS, Non-significant; **p<0.01. (E) Mean TIGIT ligand CD155 and CD112 expression from n=6 patients on tumor cells (Neuroblastoma), myeloid derived suppressor cells (MDSC), and M2 macrophages (M2) from freshly resected pediatric neuroblastoma tissue, NS, Non-significant; *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001

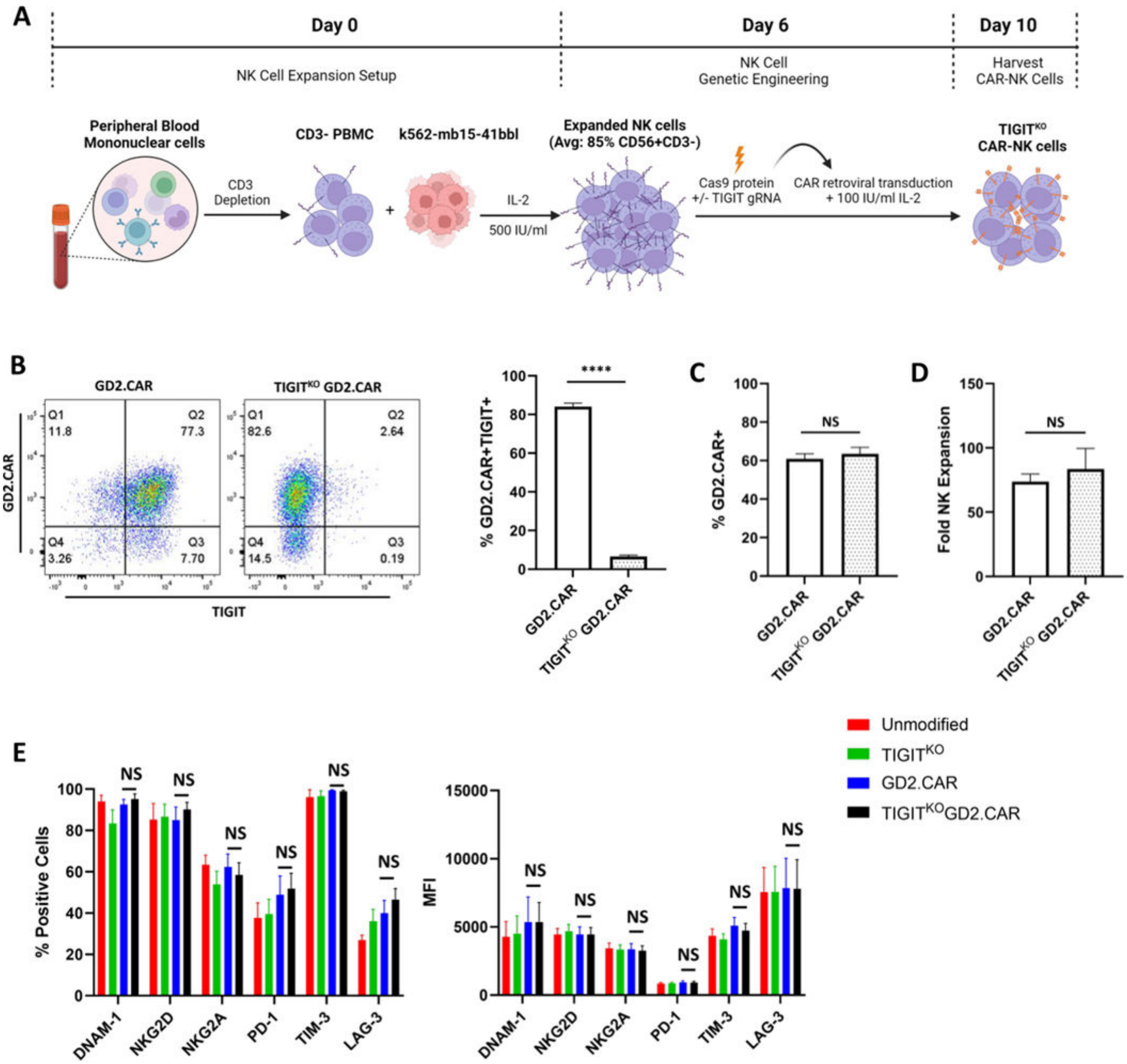

Figure 2. Genetic engineering to generate TIGIT^{KO} CAR-NKs.

(A) Protocol used for genetic modification of human expanded CD56+CD3- NK cells using sequential CRISPR/Cas9 and retroviral transduction to generate TIGIT^{KO}GD2.CAR-NKs. (B) Representative flow plots assessing CAR and TIGIT expression on GD2.CAR-NKs and TIGIT^{KO}GD2.CAR-NKs as well as Mean TIGIT expression on GD2.CAR-NKs after manufacture from n=17 healthy donors, ****p<0.0001. (C) Mean GD2.CAR expression on TIGIT-expressing and TIGIT^{KO} NK cells from n=17 healthy donors, NS, Non-significant. (D) Mean fold expansion of GD2.CAR-NKs and TIGIT^{KO}GD2.CAR-NKs after 7 days of activation using k562-mb15–41BB-L and IL-2 from n=3 healthy donors, NS, Non-significant. (E) Mean DNAM-1, NKG2D, NKG2A, PD-1, TIM-3, and LAG-3 expression on unmodified NK cells, TIGIT^{KO} NK cells, GD2.CAR-NKs, and TIGIT^{KO}GD2.CAR-NKs after manufacture from n=6 healthy donors, NS, Non-significant.

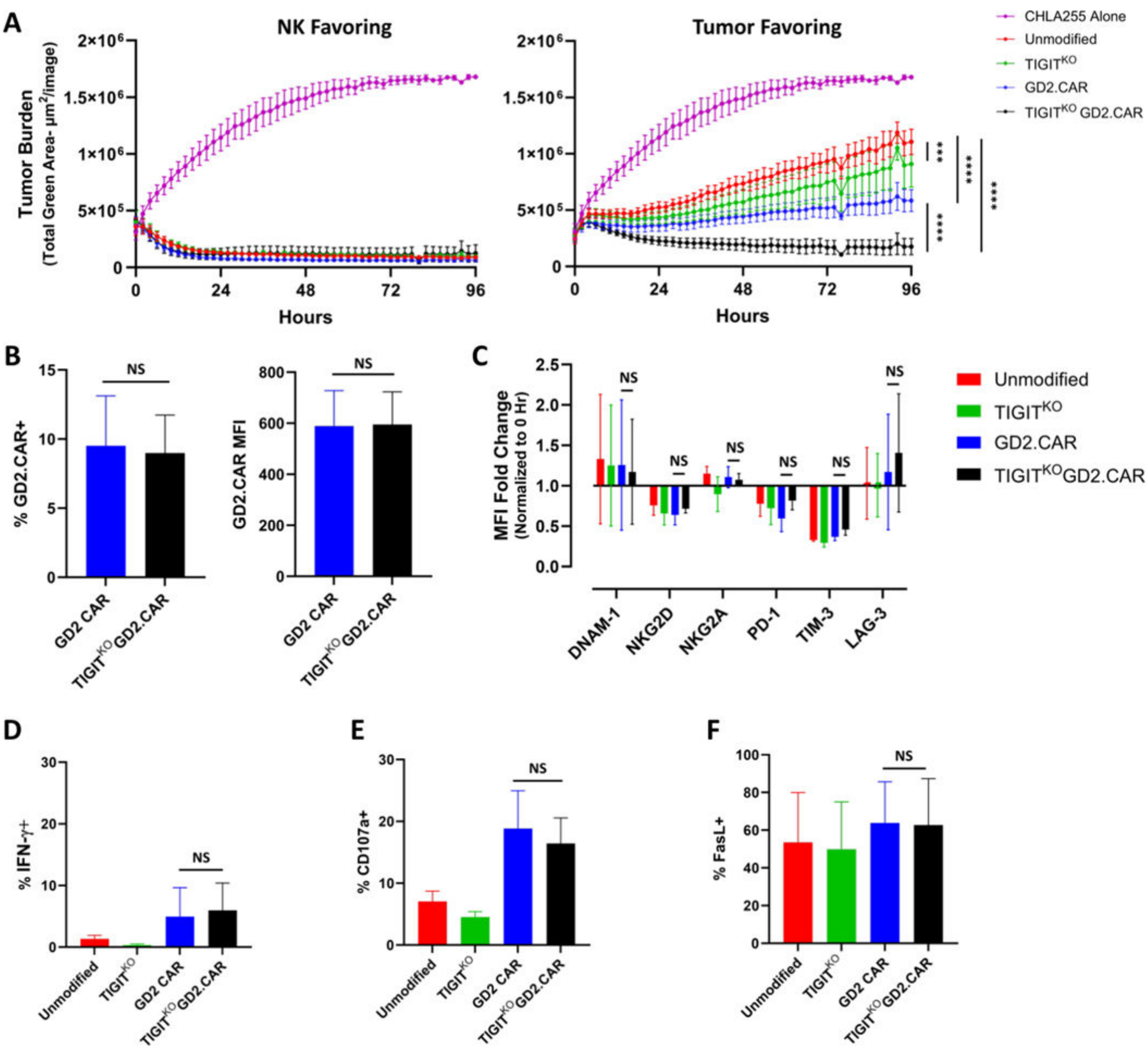

Figure 3. TIGIT^{KO} GD2.CAR-NKs exhibit enhanced tumor control in vitro.

(A) GFP-expressing CHLA255 neuroblastoma cells were co-cultured with TIGIT^{KO} GD2.CAR-NKs for 96 hours at an E:T ratio of 4:1 (NK favoring) or 1:4 (Tumor favoring). Mean (from n=4 healthy donors) tumor growth when cultured without effector (purple) or with unmodified NK cells (red), TIGIT^{KO} NK cells (green), GD2.CAR-NKs (blue), or TIGIT^{KO} GD2.CAR-NKs (black) were analyzed using IncuCyte® Area under the curve followed by One-Way ANOVA with Tukey's Multiple Comparisons Test; NS, Non-Significant, ***p<0.001, ****p<0.0001. (B) Mean GD2.CAR expression on TIGIT-expressing and TIGIT^{KO} NK cells after tumor co-culture from n=3 healthy donors, NS, Non-significant, (C) Mean fold change in DNAM-1, NKG2D, NKG2A, TIM-3, PD-1 or LAG-3 expression on unmodified, TIGIT^{KO}, GD2.CAR, and TIGIT^{KO} GD2.CAR-NKs after tumor co-culture (n=3 healthy blood donors), NS, Non-significant, (D) Mean IFN-γ expression in unmodified, TIGIT^{KO}, GD2.CAR, and TIGIT^{KO} GD2.CAR-NKs after tumor co-culture from n=3 healthy donors, NS, Non-significant, (E) Mean CD107a expression on unmodified, TIGIT^{KO}, GD2.CAR, and TIGIT^{KO} GD2.CAR-NKs after tumor co-culture from n=3 healthy donors, Non-significant, (F) Mean Fas ligand expression on unmodified, TIGIT^{KO}, GD2.CAR, and TIGIT^{KO} GD2.CAR-NKs after tumor co-culture from n=3 healthy donors, NS, Non-significant.

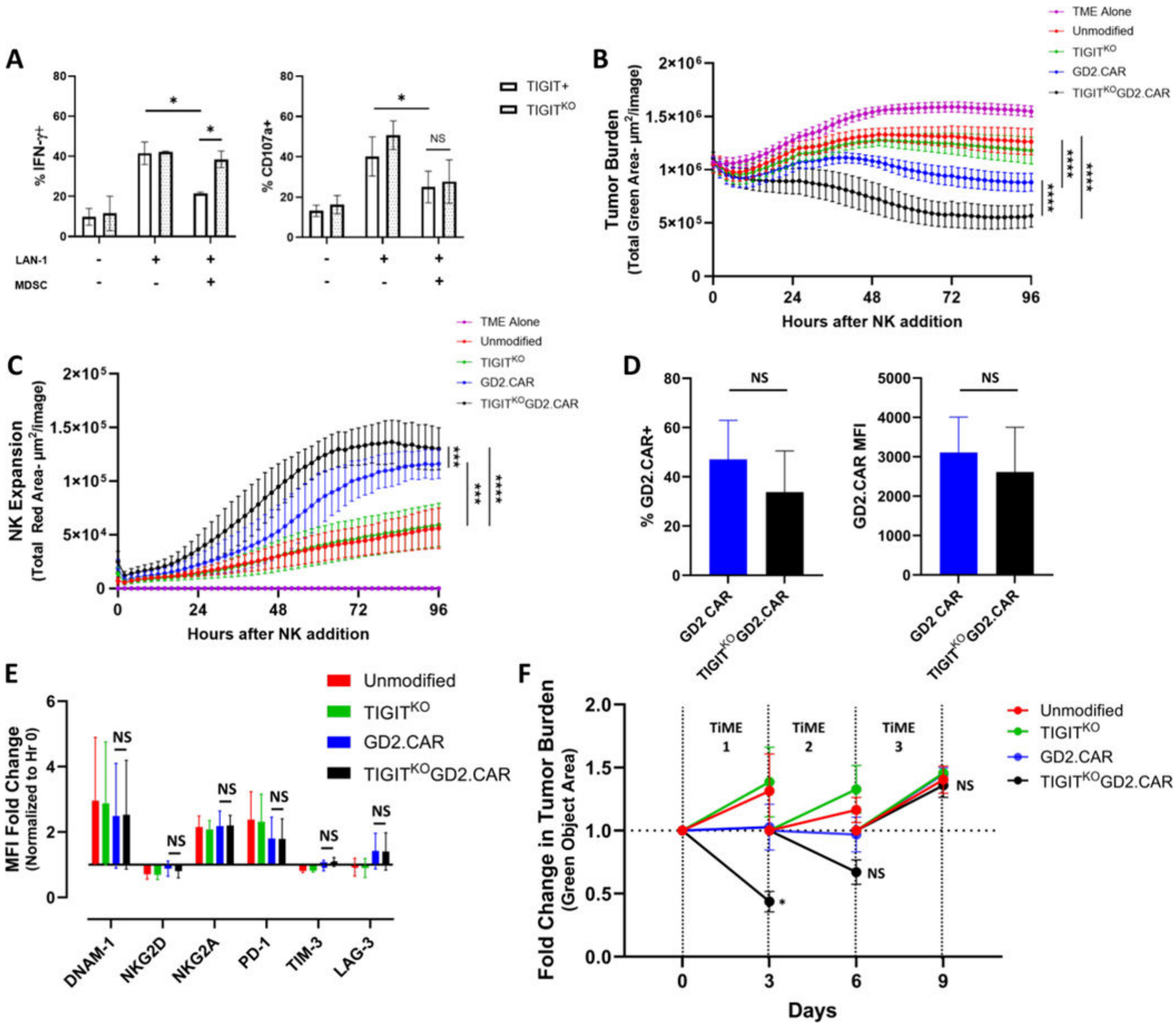

Figure 4. TIGIT^{KO} GD2.CAR-NKs expand and eliminate neuroblastoma in a TIGIT ligand-rich environment.

(A) Mean IFN-γ and CD107a expression in GD2.CAR-NKs or TIGIT^{KO} GD2.CAR-NKs cultured with LAN-1 neuroblastoma alone or in combination with MDSCs from n=3 healthy donors, NS, Nonsignificant; *p<0.05. (B) GFP-expressing CHLA255 neuroblastoma cells were cultured with CD14+ monocytes for 72 hours (TiME Establishment) prior to addition of CellTracker Red-labeled NK cells and then further co-cultured for 96 hours. Mean tumor growth and NK cell expansion kinetics (C) from n=3–4 healthy donors in co-cultures without effector (purple) or with unmodified NK cells (red), TIGIT^{KO} NK cells (green), GD2.CAR-NK (blue), or TIGIT^{KO} GD2.CAR-NK (black) were analyzed using IncuCyte®, ***p<0.01, ****p<0.0001. (D) Mean GD2.CAR expression on TIGIT-expressing and TIGIT^{KO} NK cells after TiME culture from n=4 healthy donors, NS, Non-significant. (E) Mean change in DNAM-1, NKG2D, NKG2A, TIM-3, PD-1 or LAG-3 expression on unmodified, $TIGIT^{KO}$, GD2.CAR, and $TIGIT^{KO}$ GD2.CAR-NKs after TiME culture from n=3 healthy donors, NS, Non-significant. (F) Mean fold change in neuroblastoma tumor in each of 3 sequential TiME re-challenge assays in which unmodified NK cells (red), $TIGIT^{KO}$ NK cells (green), GD2.CAR-NKs (blue), and $TIGIT^{KO}$ GD2.CAR-NKs (black) from n=3 healthy donors were repetitively challenged against established TiMEs, NS, Non-significant,*$p<0.05$.

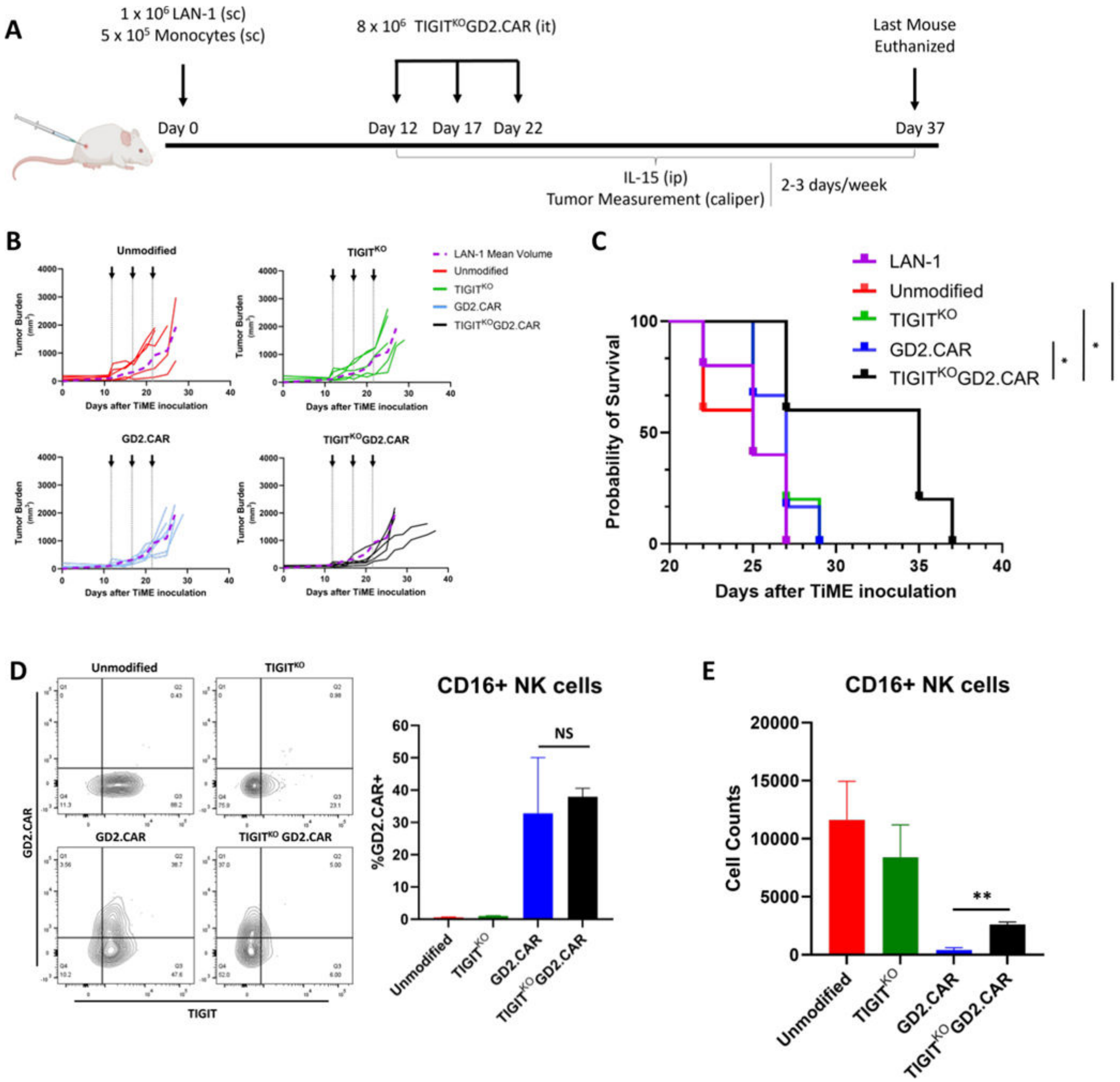

Figure 5. TIGITKO GD2.CAR-NKs extend survival in an *in vivo* TiME xenograft mouse model.

(A) Schema for neuroblastoma TiME xenograft mouse model in NSG-SGM3 mice. Mice were inoculated subcutaneously with a 2:1 mixture of LAN-1 neuroblastoma cells and CD14+ monocytes. After 12 days of tumor growth (tumors ~100 mm^3), mice were injected intra-tumorally with $TIGIT^{KO}$ GD2.CAR-NKs x3 five days apart, along with intra-peritoneal injections of 1 mg of IL-15 three times per week. (B) Spider plots of tumor burden in mice treated with unmodified (red), $TIGIT^{KO}$ (green), GD2.CAR-NK (blue), or $TIGIT^{KO}$ GD2.CAR-NK (black) compared to mean tumor burden of untreated mice (purple) (n=6 mice/group). (C) Survival curves of mice treated with unmodified (red), $TIGIT^{KO}$ (green), GD2.CAR-NK (blue), or $TIGIT^{KO}$ GD2.CAR-NK (black); *$p<0.05$. (D)

Representative flow cytometric contour plots (left) and Mean GD2.CAR expression on TIGIT-expressing and $TIGIT^{KO}$ NK cells within tumor digests harvested on day 16 (n=3 mice/group); NS, Non-significant. (E) Mean number of absolute intra-tumoral NK cells of quantified from tumor digests (n=3 mice/group); **p<0.01

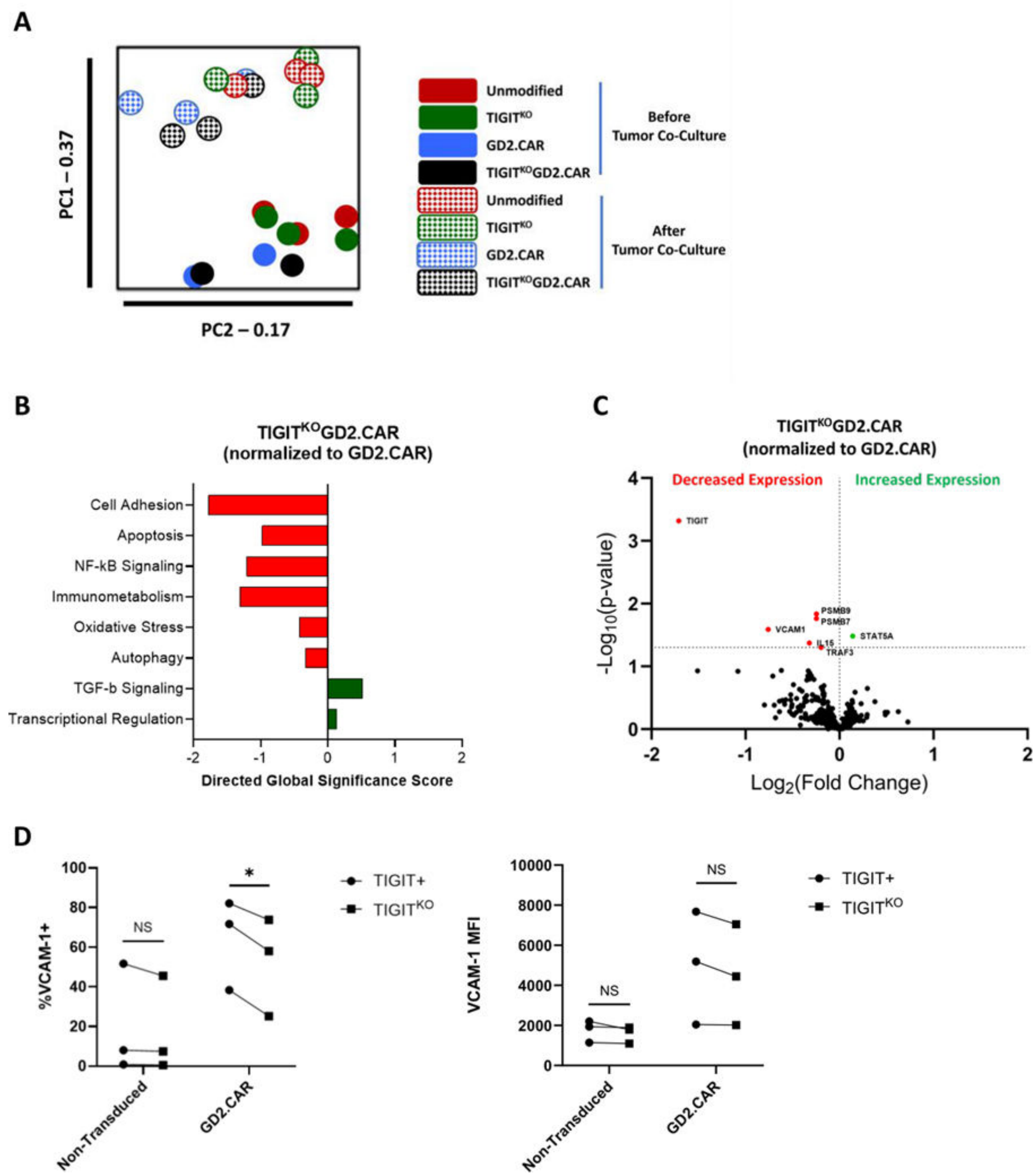

Figure 6. Transcriptome and cell adhesion analysis of TIGITKO GD2.CAR-NKs after tumor co-culture.

(A) RNA isolation of NK cells was performed before and after 72-hour tumor co-culture and transcriptomic changes quantified using NanoString. Principal component analysis of gene expression in unmodified, TIGIT^{KO}, GD2.CAR-NK, and TIGIT^{KO}GD2.CAR-NKs before and after co-culture. (B) Volcano plot analysis of TIGIT^{KO} GD2.CAR-NK gene expression compared to control GD2.CAR-NK cells after tumor exposure (n=3 healthy blood donors). (C) Gene ontology of TIGIT^{KO} GD2.CAR-NK gene expression compared to control GD2.CAR-NKs after tumor exposure (n=3 healthy blood donors). (D) %

VCAM-1 and mean fluorescence expression on unmodified, TIGIT^{KO}, GD2.CAR-NK, and TIGIT^{KO}GD2.CAR-NKs after 72-hour CHLA255 tumor co-culture (n=3 healthy blood donors); NS, Nonsignificant; *p<0.05.

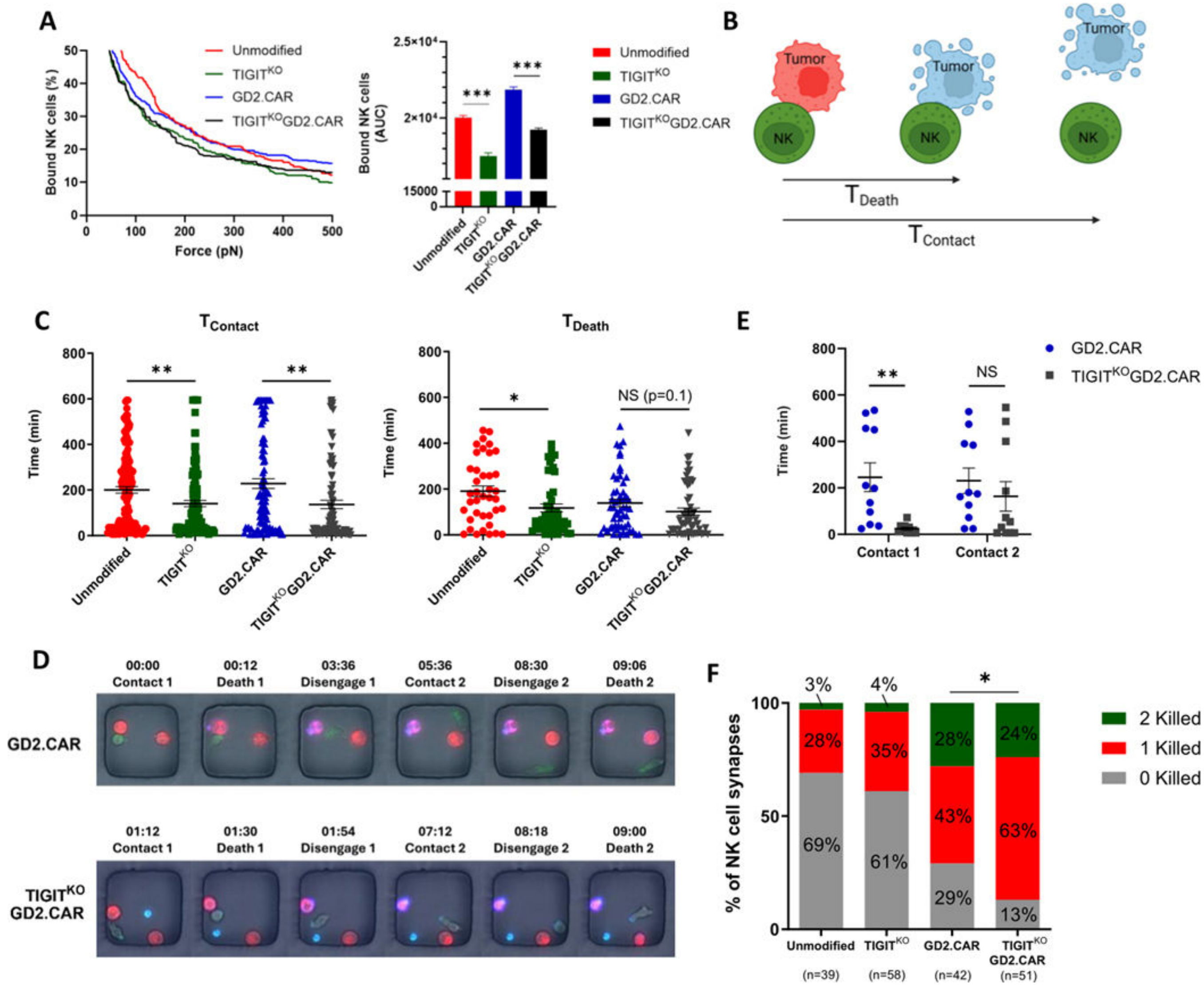

Figure 7. TIGIT deletion reduces GD2.CAR-NK cell synapse duration with tumor targets

(A) Mean number of bound NK cells to CHLA255 tumor targets to measure cell avidity of unmodified, TIGIT^{KO}, GD2.CAR-NK, and TIGIT^{KO} GD2.CAR-NKs to target cells (n=3 replicates); ***p<0.001. (B) Schematic representation for TIMING assay. Unmodified, TIGIT^{KO}, GD2.CAR-NK, and TIGIT^{KO}GD2.CAR-NKs (PKH26 Green cell membrane label); CHLA255 tumor cells (PKH67 Red cell membrane label) at 1E:1T and 1E:2T effector:target ratios. Annexin V (blue) marker for tumor cell death. (C) Duration of synapse ($T_{Contact}$) and time to tumor death (T_{Death}) of all encounters in nanowells containing a 1E:1T ratio (n=91–143 single cells), NS, Non-significant; *p<0.05, **p<0.01 (D) Representative micrographs of GD2.CAR-NK (Top Panel), TIGIT^{KO}GD2.CAR-NK (Bottom Panel), interacting with CHLA255 tumor cells. (E) Duration of individual synapse times of GD2.CAR-NK and TIGIT^{KO}GD2.CAR-NKs with CHLA255 from nanowells with 1E:2T ratios (n=10–11), NS, Non-significant; **p<0.01. (F) Proportion of unmodified,

$TIGIT^{KO}$, GD2.CAR-NK, and $TIGIT^{KO}$ GD2.CAR-NKs that killed zero, 1, or 2 tumor targets from nanowells with a 1E:2T ratio (n=39–58 replicates), *p<0.05.